

Lecture 07: Optimal Labor Taxation – Live In-Class Empirical Exercise

Stata Code: Revenue Maximizing Rates (Live In Class)

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* Optimal Labor Tax Exercises - Undergrad-Friendly, No preserve/restore
* Tasks:
*   1) Full-sample labor Laffer curve
*   2) Top earners (>= $800k): revenue decomposition + top-rate Laffer
*   3) Empirical Pareto  $a(z^*)$  across income cutoffs
* Data: usdina2020.dta (PSZ/DINA). Weights scaled by 100,000.
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*-----
* Paths and parameters (globals for easy reuse)
*-----
global ROOT "${dropbox}/LectureEmpiricalExercises/
    Lecture07_optlabtax_ch2021/liveinclass"
global DATA "$ROOT/usdina2020.dta"
global FIGS "$ROOT"
global TAXGRID "0_0.02_0.04_0.06_0.08_0.10_0.12_0.14_0.16_0.18_0.20_0.22_
    0.24_0.26_0.28_0.30_0.32_0.34_0.36_0.38_0.40_0.42_0.44_0.46_0.48_0.50_
    0.52_0.54_0.56_0.57_0.58_0.60_0.62_0.64_0.66_0.67_0.68_0.70_0.72_0.74_
    0.76_0.78_0.80_0.82_0.84_0.86_0.88_0.90_0.92_0.94_0.96_0.98_1"
global ZSTAR 622051 // 2020 threshold for married filing jointly
622050
global ETI_BASE 0.50
```

Stata Code: Optimal Tax Rate Simulations

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    0.76_0.78_0.80_0.82_0.84_0.86_0.88_0.90_0.92_0.94_0.96_0.98_1"
global ZSTAR 622051 // 2020 threshold for married filing jointly
622050
global ETI_BASE 0.50
global income_variable1 fiwag // fiscal income wages abd
pensions,
global income_variable1 fiinc // fiscal income including capital
gains
global income_variable2 flinc // personal factor labor income,
global income_variable3 peinc // pre-tax national income
global tax_variable1 ditax + govcontrib
global tax_variable2 ditaf + ditas + ssuicontrib + othercontrib
global tax_variable3 tax

*-----
* Build base sample once and save
*-----
* LOAD WORKING-AGE PEOPLE
use "$DATA" if id>0, clear
bys id: egen adults_work = mean(inrange(age,30,55))
keep if adults_work==1
keep if married==1
bys id: keep if _n==1

* income, taxes, weights
gen income = fiinc
gen taxes = ditax + govcontrib
gen w = dweghttaxu/100000

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label var income "Income_($)"
label var taxes "Taxes_($)"
label var w "Population_weight"

* REMOVE WEIRD OBSERVATIONS
gen t0 = taxes/income
keep if income>0 & t0>0 & t0<0.99

save "$ROOT/base_sample.dta", replace

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* 1) CURRENT AVERAGE TAX RATES AND TOP-BRACKET PARETO PAREMETER
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use "$ROOT/base_sample.dta", clear

*CURRENT AVERAGE TAX RATES (WEIGHTED BY PEOPLE)
sum t0 [weight=w]
sum t0 [weight=w] if income>650000

*TOP 1% PARETO PARAMETER
keep if income >= 650000
tabstat income [aw=w], stat(mean)
disp 1861301/(1861301-650000)
local a_top = 1861301/(1861301-650000)

*WHAT THE REVENUE-MAXIMIZING LINEAR RATE FOR THE FULL POPULATION SHOULD BE
disp 1/(1+.5)

*WHAT THE REVENUE-MAXIMIZING RATE FOR THE TOP 1% SHOULD BE
disp 1/(1+1.54*.5)
local t_opt = 1/(1 + 'a_top'*$ETI_BASE) // optimal rate using a(z*) and
assumed ETI

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* 2) FULL-SAMPLE LABOR LAFFER CURVE
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use "$ROOT/base_sample.dta", clear
gen eti = 0.5

* revenue under each candidate rate
local i = 1
foreach t of global TAXGRID {
    gen Revenue'i' = 't' * income * ((1-'t')/(1-t0))^eti * w
    local ++i

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}

* total revenue by rate
collapse (sum) Revenue*
set obs 200
gen tax_rate = .
gen total_revenue = .
label var tax_rate "Labor_Tax_Rate(%)"
label var total_revenue "Revenue(trillions)"

local i = 1
foreach t of global TAXGRID {
    replace tax_rate = 't' in 'i'
    replace total_revenue = Revenue'i'[1] in 'i'
    local ++i
}

keep if total_revenue < .
replace total_revenue = total_revenue/1e12
replace tax_rate = round(tax_rate*100)

summ total_revenue, detail
local max_rev = r(max)
gen laffer_rate = tax_rate if total_revenue >= 'max_rev' - .000000001
summ laffer_rate if laffer_rate < .
local laffer_star = r(mean)
local label_x = 'laffer_star' - 7
local label_y = 'max_rev'*0.9

twoway line total_revenue tax_rate, lwidth(thick) lcolor(navy) ///
    ytitle("Total_Labor_Tax_Revenue(Trillions)") ///
    xtitle("Labor_Tax_Rate(%)" ) ///
    xlabel(0(10)100, labsize(medlarge)) ///
    xline('laffer_star', lpattern(dash)) ///
    text('label_y' 'label_x' "Laffer_Rate='laffer_star'%", placement(top)
        size(small)) ///
    title("Laffer_Curve:_Labor_Tax_Revenue") ///
    graphregion(color(white))

graph export "$FIGS/laffer_curve.png", replace as(png)

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* 3) TOP-RATE LAFFER
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use "$ROOT/base_sample.dta", clear
global ZSTAR=650000
gen base_top = max(income - $ZSTAR, 0)
gen t0_top = t0
gen rev_fixed = taxes - t0_top*base_top

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quietly summarize income [aw=w] if income > $ZSTAR, meanonly
local zm = r(mean)
local a = 'zm'/'(zm' - $ZSTAR)
di "Empirical_Pareto_a(z*)_for_top_bracket:_'a'"
local eti_top = $ETI_BASE*'a'

local i = 1
foreach t of global TAXGRID {
    * behavioral response for top base using Saez formula on net-of-tax rate
    gen Revenue'i' = (rev_fixed ///
        + 't' * base_top * ((1-'t')/(1-t0_top))~'eti_top') * w
    local ++i
}

collapse (sum) Revenue*
set obs 200
gen tax_rate = .
gen total_revenue = .

local i = 1
foreach t of global TAXGRID {
    replace tax_rate = 't' in 'i'
    replace total_revenue = Revenue'i'[1] in 'i'
    local ++i
}

keep if total_revenue<.
replace total_revenue = total_revenue/1e12
replace tax_rate = round(tax_rate*100)

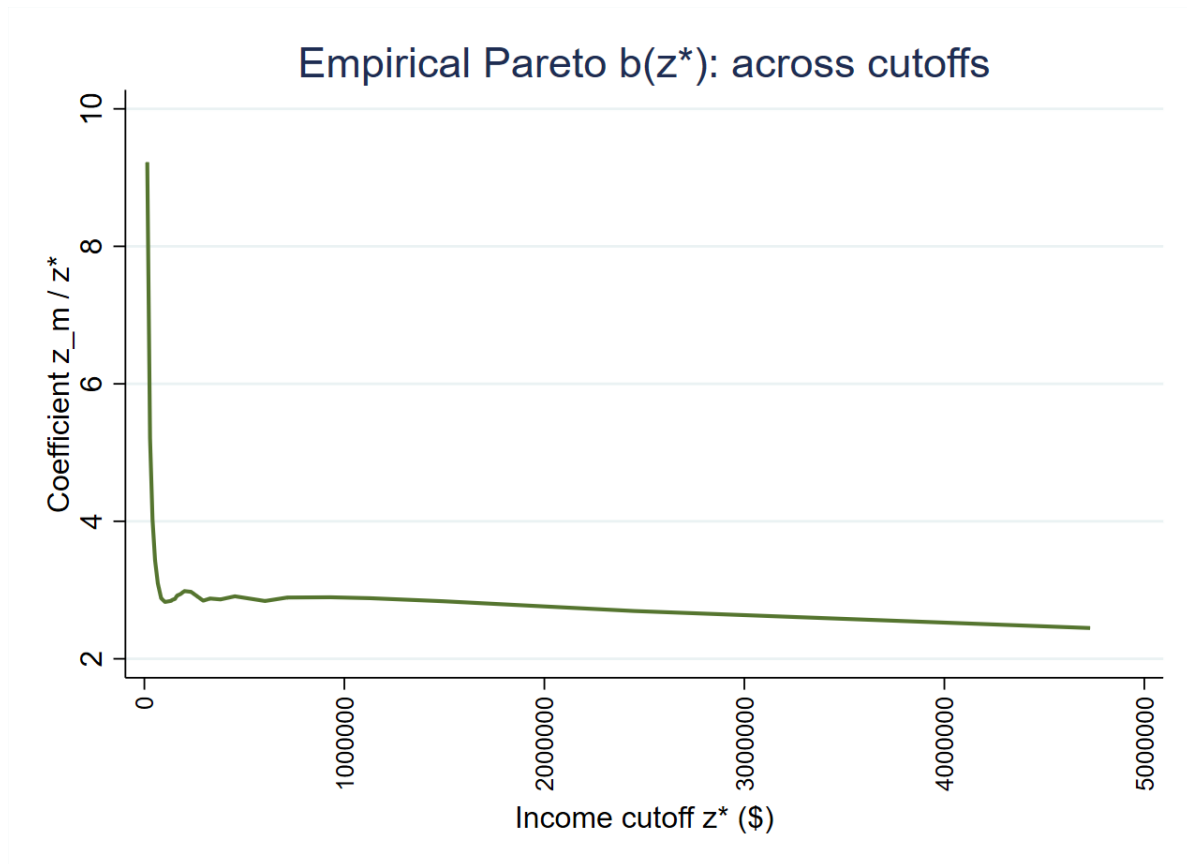
summ total_revenue, detail
egen max_rev_top = max(total_revenue)
* Create the laffer rate
gen laffer_rate = tax_rate if total_revenue >= max_rev_top
summ laffer_rate if laffer_rate<.
local laffer_star_top = r(mean)
summ max_rev_top, detail
local max_rev_top = r(mean)
local label_x_top = 'laffer_star_top' - 7
local label_y_top = 'max_rev_top'*0.9

line total_revenue tax_rate, lwidth(thick) lcolor(navy) ///
ytitle("Total_Labor_Tax_Revenue_(Trillions)") ///
xtitle("Top_Labor_Tax_Rate_above_$800k_(%)") ///
xlabel(0(10)100, labsize(medlarge)) ///
xline('laffer_star_top', lpattern(dash)) ///
text('label_y_top' 'label_x_top' "Laffer_Rate='laffer_star_top'%",
    placement(top) size(small)) ///
title("Laffer_Curve:_Change_only_top_rate") ///
graphregion(color(white))

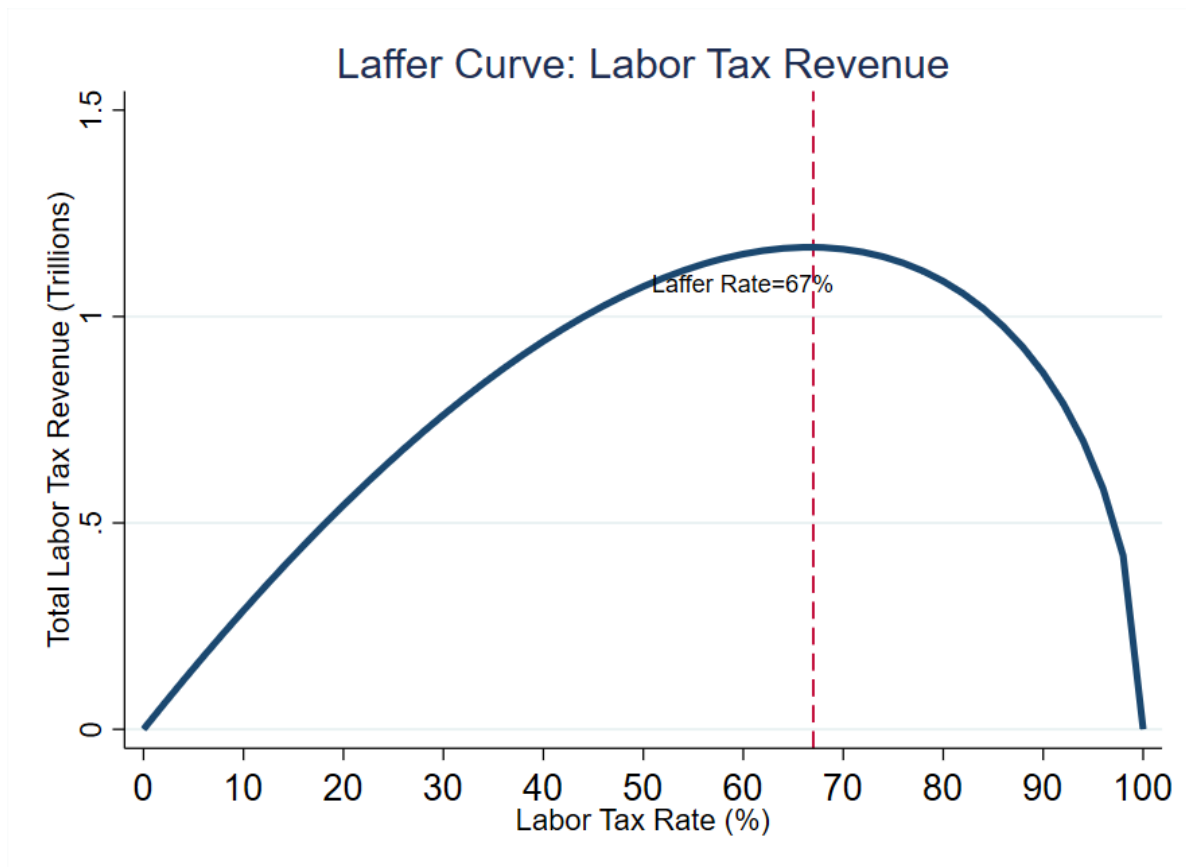
graph export "$FIGS/laffer_curve_toprate_800k.png", replace as(png)

```

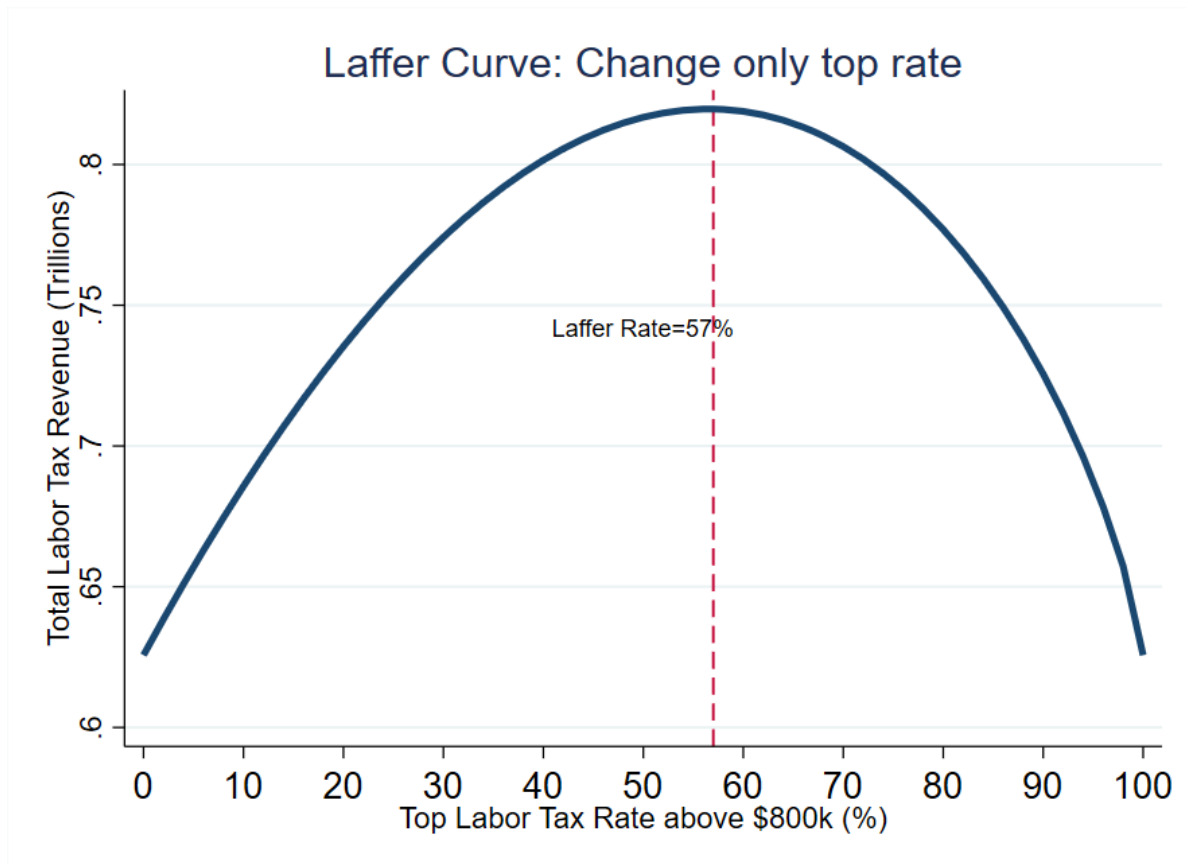
Results: Pareto Parameter by Income



Results: Laffer Curve



Results: Laffer Curve for Top Rate at \$800k



Results: Labor 800k Tax Rate and Revenue Decomposition

